<u>STATEMENT OF WORK</u>
<u>10X10 Optimus Pressure Systems Hardware</u>

1. Objective/Requirements

The10x10 Supersonic Wind Tunnel (SWT) is currently upgrading the facility Pressure Measurement System to the TE Optimus Data System designed specifically for wind tunnel testing. The 10x10 SWT at NASA GRC is standardized to using TE Connectivity/ Measurement Specialties products for pressure measurement. The system currently in use (ESP), is the predecessor of the Optimus Pressure Measurement System and uses the same interface to connect with the NASA data system infrastructure. The interface between the Optimus system and NASA's data system has been developed over the last decade and is well understood, being in use at various other facilities at NASA GRC.

2. Characteristics, Scope, and Specs

The Optimus Pressure System shall support up to sixty-four (64) individual miniature pressure scanners of various ranges during 10x10 Wind Tunnel Testing. The Optimus Data System will require the following cables in order to complete the current hardware installation for checkout and future testing.

System shall require the following cables to be purchased:

QTY	Description	Specifications
5	**<u>8476-0010000001</u>** **<u>MSDI power cable</u>**	Able to provide regulated power to TE Connectivity/ Measurement Specialties MSDI, miniature scanner digitizing interface.
50	**<u>OSCB-0000000003</u>** **<u>Scanner cable</u>**	Able to connect a single 32HD or 64HD miniature pressure scanner to MSDI miniature scanner digitizing interface. Able to send regulated power to a single 32HD or 64HD miniature pressure scanner. Able to send analog signals back to the MSDI from a single 32HD or 64HD miniature pressure scanner. Able to carry digital channel selection signal to a single 32HD or 64HD miniature pressure scanner.

3. **Place of Performance**
   NASA Glenn Research Center
   21000 Brookpark Rd
   Cleveland OH 44135-3127

4. **Warranty**
   The vendor shall provide a warranty against defects in materials and workmanship for a period of one (1) year following completion of work. Any nonconformities identified by NASA as defects in materials and workmanship from the original delivery shall be handled as Non-Conforming Deliverables as described above.

5. **Invoices**
   In accordance with commercial practice, software licenses and software, hardware maintenance are provided as non-severable services. Software maintenance includes the development and publishing of "bug" and/or defect fixes, patches, updates, upgrades in function, technology to maintain the operability and usability of the software product, as well as various technical support and diagnostic tools and resources. Hardware maintenance includes the labor for troubleshooting and part replacement.

   Payment will be made upon completion of tasks or deliverables. Invoices shall include the following: task order/delivery order number/contract number, itemized details of the individual costs and the total invoice cost.